



ADAPTIVE MOBILITY

Behavioral Simulations in Urban Traffic Systems



1 PROBLEM

Traffic ≠ Infrastructure Problem

- Congestion persists despite more roads
- Systems assume rational behavior
- Reality: decisions are dynamic, emotional and context-driven



The issue is not the roads, it's how people behave in the system.

2 WHY CURRENT APPROACHES FAIL

The Old Map Doesn't Work Anymore

- Static pricing → no real-time adaptation
- Models use averages, not real people
- Ignore behavioral diversity and feedback loops



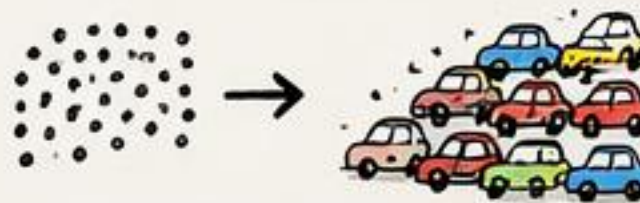
We are modeling traffic, not human behavior.

3 CORE INSIGHT

Behavior Drives the System



- People respond differently to: price, time, habits, information
- Small individual decisions...



...lead to big system effects.

Traffic patterns emerge from thousands of individual choices.

4 OUR SOLUTION

ADAPTIVE BEHAVIORAL PRICING SYSTEM

1. Behavioral Segmentation



Group travelers by: price sensitivity, time sensitivity, habits

2. Simulation Layer (Agent-Based Model)



Simulate individual decisions and test multiple scenarios

3. Dynamic Pricing Engine



Real-time pricing adjusts based on predicted behavior and system state

4. Feedback Loop



Learn from outcomes
Adjust. Improve. Repeat.

EXAMPLE SCENARIO

Peak hour pricing +20%



Price-sensitive users shift time / route



Congestion redistributed



System efficiency improves



Small nudges. Smarter choices. Better cities. 😎

5 IMPACT

From Reactive to Proactive Urban Mobility



Less Congestion



Higher Efficiency



Better Equity & Accessibility



Smarter Cities

Better decisions today. Better cities tomorrow! ❤️



CAN DYNAMIC PRICING SHAPE BEHAVIOR? **YES.** CAN IT OPTIMIZE THE SYSTEM? WE CAN **SIMULATE** IT. WE CAN **CONTROL** IT. WE CAN **IMPROVE** IT.